# MTH2010

## UNIVERSITY OF EXETER

## FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

## MATHEMATICS AND STATISTICS

### January 2024

**Groups, Rings and Fields**

**Module Leader: Mohamed Saïdi**

**Duration: 2 HOURS**

**Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

No Word Count specified

## SECTION A

**All rings are commutative with an identity element** $1$ **and** $1 \neq 0$**.**

1. (a) Let $G$ be a group and $H$, $K$ subgroups of $G$. Prove that $H \cup K$ is a subgroup of $G$ if and only if either $H \subset K$ or $K \subset H$. (10)

   (b) Let $D_{10}$ be the dihedral group with 10 elements.

   (i) Find the conjugacy classes of $D_{10}$ and write down the class equation for $D_{10}$. (10)

   (ii) Find the centre of $D_{10}$. (5)

   (c) Let $R$ be a ring.

   (i) Define what it means for a nonempty subset $I$ of $R$ to be an ideal, and a prime ideal. (8)

   (ii) Let $\phi : R \to S$ be a ring homomorphism. Define what the kernel $\mathrm{Ker}(\phi)$ of $\phi$ is and show that $\mathrm{Ker}(\phi)$ is an ideal of $R$. (10)

   (iii) Assume that $S$ is an integral domain. Show that $\mathrm{Ker}(\phi)$ is a prime ideal. (7)

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## SECTION B

**All rings are commutative with an identity element** $1$ **and** $1 \neq 0$**.**

2. Let $G$ be a group.

   (a) Let $N$ be a subgroup of $G$ contained in the centre $Z(G)$ of $G$. Show that $N$ is a normal subgroup of $G$. (7)

   (b) Show that if the quotient group $G/Z(G)$ is cyclic then the group $G$ is abelian. (10)

   (c) Assume $G$ is finite, $H$ is a subgroup of $G$, and $N$ is a normal subgroup of $G$. Let $|H|$ be the cardinality of $H$ and $|G : N|$ the index of $N$ in $G$. Show that if $\gcd(|H|, |G : N|) = 1$ then $H$ is contained in $N$. (8)

   [**25**]

3. (a) Let $S_4$ be the group of permutations of the set $\{1, 2, 3, 4\}$. Show that there are no surjective homomorphisms from $S_4$ to a group of cardinality 3 or a group of cardinality 8. (10)

   (b) A ring $R$ is called a *local ring* if it has a unique maximal ideal.

      (i) Prove that if $R$ is a local ring with maximal ideal $M$ then every element of $R$ which doesn't belong to $M$ is a unit. (You can use without proof the fact that every ideal $I$ of $R$, not equal $R$, is contained in a maximal ideal of $R$.) (8)

      (ii) Let $R$ be a ring and $M$ the set of elements of $R$ which are not units. Suppose that $M$ is an ideal of $R$. Show that $R$ is a local ring with unique maximal ideal $M$. (7)

   [**25**]

4. (a) Let $\mathbf{Z}$ be the ring of integers,

$$R = \{f(X) = a_0 + a_1X + \ldots + a_{n-1}X^{n-1} + a_nX^n : n \geq 0, a_0, a_1, \ldots, a_n \in \mathbf{Z}\}$$

the ring of polynomials with coefficients in $\mathbf{Z}$, and

$$I = (5, X)_R = \{5f(X) + Xg(X) : f(X), g(X) \in R\}$$

the ideal of $R$ generated by 5 and $X$.

By constructing a surjective ring homomorphism from $R$ to a field $F$ with kernel $I$, or otherwise, show that $I$ is a maximal ideal of $R$. Clearly state any theorems from the lecture you use. (13)

(b) Let $G$ be a group and $H$ a subgroup of $G$. Let $p$ be a prime number which divides the cardinality of $H$. Let $N$ be a normal $p$-Sylow subgroup of of $G$. Show that $N \cap H$ is the unique $p$-Sylow subgroup of $H$. (12)

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